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ROBOTIC CASING STABBING GUIDE

Abstract

A robotic casing or pipe stabbing guide for oil and gas well drilling includes a mount adapted to connect to a robotic arm, a main body, arms, casing guides, shoes and springs biasing the arms in a closed position around a casing guide. The casing or pipe guide includes a portion adapted to hold a box end of a tubular and a portion adapted to align and direct a pin-end of a second tubular into the box end of the first tubular. The arms are then moved outwardly away from the first and second tubular thus allowing for easy release. The main body assembly includes a tri-guide design adapted to better grip and control the tubular during the mounting of the second tubular.

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Background/Summary

TECHNICAL FIELD

[0001] The invention generally relates to oil and gas drilling rig equipment and more particularly to equipment used on robotic assisted drilling rigs to assist in guiding pipe when installing joints of pipe into the oil or gas well bores.

BACKGROUND

[0002] Generally, a stabbing guide is a two-piece construction hand operated tool used in the oil and gas industry to help operators align two separate sections of drill pipe, tubing, or casing for male-to-female connections. The two-piece construction of the stabbing guide is a limitation that can interfere with its effective operation. Using the right stabbing guide is the way to prevent pipe and thread damage and minimize potential connection failure and missed stabbing procedure. A good stabbing guide will help operators save time and ensure no thread damage while stabbing.

[0003] When a pipe is mis-stabbed, downtime increases, and there is potential damage to the thread connection. To help increase efficiency and reduce risk to operators, robotic casing stabbing guides were developed for the oil and gas well drilling industry for a semi-autonomous drilling rig.

However, robotic systems have their own limitations and do not properly operate in all conditions. The use of robotic arms on a drilling rig's floor prevents workers from entering onto the floor to install and remove casing guides between each casing segments on the rig.

[0004] In well drilling, specifically including oil and gas well drilling, conventional casing operations typically involve specialized crews and equipment that are brought to a rig site for the sole purpose of running the casing. The casing crew rigs up tools and operates the equipment, connecting each joint of casing to be lowered into the well bore. The driller who operates the topdrive will use a stab-in procedure of the casing running tools. The use of stabbing guides into each joint of the box end on the casing is done to help entry, but use of sensors on the drilling platform halts operation when a person sets foot on the rig's floor. As a result, manual use of stabbing guides is unavailable. The use of alternatives would make this process more efficient

[0005] A solution is needed that is capable of operating in harsh weather conditions such as in -40° C. weather. The solution must interface with the onboard robotic end effector and be devoid of hydraulic and electrical systems. The solution needs to adapt to the different casing sizes including casing diameters, allow for mounting a doping unit, allow dope and steam to be sprayed on the casing during operation, and for the robotic manipulator to remove the solution from around the casing after connection.

SUMMARY

[0006] It is an objective of the invention to offer a solution to the above-described problems. The invention offers a better solution in a robotic environment to allow oil or gas well rig operators to align two separate sections of drill pipe, tubing, or casing for male-to-female connections. It should be understood that although reference is frequently given to using the invention with oil or gas well casings, the invention is intended to be used with drill pipe, tubing, or casings. The invention will work with robotic systems including fully-automated advanced system robotics for performing a variety of tasks and improving a rig's performance. With the use of robotics and the removal of personnel on a rig floor there is a need for this invention.

[0007] In a nonlimiting embodiment, the robotic stabbing guide of the invention is constructed of three pieces, for a smaller footprint used to help align $9\frac{5}{8}$ inch to $2\frac{3}{8}$ inch pipe and protect the thread and sealing area of the connection. The stabbing guide of the invention can be bolted onto any rig that uses robotic assisted pipe handlers.

[0008] In a nonlimiting embodiment of the invention, a robotic casing stabbing guide for oil and gas well drilling, includes a mount, adapted to connect to a robotic arm, and a main body associated

with the mount, centering springs. The main body includes a cylindrical casing guide including a top portion having an inwardly angled surface relative to an axis passing through the center of the cylindrical casing guide, and a bottom portion having an inner surface colinear with the axis. The main body also includes a first and second arm in pivoting association with the main body wherein the first and second arm are biased into a closed position by at least one biasing element, wherein the first arm biases a first section of the cylindrical casing guide and the second arm biases a second section of the cylindrical casing guide, wherein the first arm includes a first shoe portion curved and extending outwardly away from the cylindrical casing guide, wherein the second arm includes a second shoe portion curved and extending outwardly away from the cylindrical casing guide.

[0009] In a nonlimiting embodiment of the robotic casing stabbing guide the cylindrical casing guide is in three segments. In another nonlimiting embodiment the three segments are each 120° about the axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Advantages of the present invention will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

[0011] FIG. 1A is a top plan view of a nonlimiting embodiment of a robotic casing stabbing guide apparatus of the invention;

[0012] FIG. 1B is a side plan view of the apparatus of FIG. 1A;

[0013] FIG. 1C is an exploded view of the apparatus of FIG. 1A;

[0014] FIG. 1D is a rear perspective view of the apparatus of FIG. 1A;

[0015] FIGS. 2A-2B are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0016] FIGS. 3A-3B are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0017] FIGS. 4A-4B are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0018] FIGS. 5A-5B are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0019] FIGS. 6A-6B are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0020] FIGS. 7A-7C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0021] FIGS. 8A-8C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0022] FIGS. 9A-9C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0023] FIGS. 10A-10C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0024] FIGS. 11A-11C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0025] FIGS. 12A-12C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0026] FIGS. 13A-13C are plan views of a part of the apparatus of FIG. 1A, or an embodiment thereof;

[0027] FIGS. **14A-14C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0028] FIGS. **15A-15C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0029] FIGS. **16A-16D** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0030] FIGS. **17A-17C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0031] FIGS. **18A-18C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0032] FIGS. **19A-19C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0033] FIGS. **20A-20C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0034] FIGS. **21A-21C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0035] FIGS. **22A-22C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0036] FIGS. **23A-23D** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0037] FIGS. **24A-24D** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0038] FIGS. **25A-25C** are plan views of a part of the apparatus of FIG. **1A**, or an embodiment thereof;

[0039] FIG. **26A** is a top view of the apparatus of FIG. **1A** in an open configuration;

[0040] FIG. **26B** is a front perspective view of the apparatus of FIG. **1A** in an open configuration;

[0041] FIG. **26C** is a rear view of the apparatus of FIG. **1A** in an open configuration;

[0042] FIGS. **27A-27D** illustrates an example process of opening and closing the apparatus of FIG. **1A** around a tubular;

[0043] FIG. **28** illustrates an example process for utilizing the apparatus of FIG. **1A** to secure a tubular during installation; and

[0044] FIG. **29** illustrates an example process for utilizing the apparatus of FIG. **1A** to secure a tubular during uninstallation.

DETAILED DESCRIPTION

[0045] Referring to FIGS. **1-25**, in a nonlimiting embodiment of the invention a robotic tubular stabbing guide **10**, is a stabbing guide adapted to mount to a robotic arm end effector (not shown), with mounting points for a doper. Generally, a robotic end effector is a peripheral device that attaches to a robot's arm, allowing the robot to interact with its task. Robotic systems including robotic arm end effectors have recently been introduced in the gas and oil well drilling industry. The robotic tubular stabbing guide **10** is intended to be able to associate with any of the robotic systems used in the oil and gas well industry including their robotic arm end effectors.

[0046] The robotic tubular stabbing guide **10** includes a cylindrical polyurethane guide intended to rest on a previously installed length of a casing's box end. A top portion of the polyurethane guide is angled relative to an axis passing through the center of the cylindrical polyurethane guide. A bottom portion of the polyurethane guide is colinear with the axis. The bottom portion of the polyurethane guide holds the casing or pipe box end while the top portion of the polyurethane guide directs the pin end of the next joint of casing or pipe into the center of the box end. As the pin end enters the box end through the direction of the polyurethane guide, the polyurethane guide holds the pipe in its 360° circumference so that the pin end contacts and slides along the angled wall of the top portion of the polyurethane guide. Thus, the robotic tubular stabbing guide **10** keeps

the tubular threads, shoulders, and seals from being damaged. Once a rigid connection is made, the robotic tubular stabbing guide **10** is capable of being pulled free from the casing, by splitting along seams in its diameter. The three-piece design is used to keep the opening footprint to the smallest physical size. The robotic tubular stabbing guide **10** uses the force of the robotic arm to pull free from the casings/pipes. Springs along the RCSG's **10** joints keep the joints closed if pipe is misstabbed and when not being pushed or pulled against the tubular diameter including if the above joint had a missed stab. Shoes along the front of the robotic tubular stabbing guide **10** allow it to be opened by pressing against the diameter of the casing. Geometry directs the force of this interaction to optimally open the robotic tubular stabbing guide **10**. The need for this to be pushed on to the pipe if the connection failed to make up properly and the joint must be broke out and inspect the threads. The robotic tubular stabbing guide **10** also features some vertical travel, to accommodate misalignment with the casing stump height relative to the robotic end effector.

[0047] The robotic tubular stabbing guide **10** consists of a mount **12** including centering spring assembly, a main body assembly **14**, a first arm assembly **16**, a second arm assembly **18**, a first casing guide, **20**, a second casing guide **22**, a third casing guide **24**, a first shoe assembly **26**, a second shoe assembly **28**, and springs **30**. The mount **12** includes a mounting plate **32** with a bolt pattern **34** adapted to interface with an end effector of a robotic arm, and two rails **36** slidably associated with a centering spring assembly **35**, the centering spring assembly **35** being movably associated with the main body assembly **14**. In a nonlimiting embodiment, the rails **36** are associated with the centering spring assembly **35** so that the main body assembly **14** may move vertically up and down along the rails **36** within a limited range of travel. The centering spring assembly **35** includes a bolt, spring and nut assembly **37** adapted to center and align the robotic tubular stabbing guide **10** as attached to a robotic arm to better grasp a pipe or casing being worked with in a stabbing process. It should be appreciated that the mount **12** and bolt pattern **34** are illustrative of a way to associate the robotic tubular stabbing guide **10** to the robotic end effector, it being understood that persons of ordinary skill in the art could use many different well-known ways of associating the robotic tubular stabbing guide **10** to the end effector.

[0048] The main body assembly **14** includes a tri-guide design **40**, adapted to better grip and control a tubular during the mounting of a second tubular. The tri-guide design **40** includes the first casing guide **20**, the second casing guide **22**, and the third casing guide **24**, that are adapted to pivot at the first pivot point **42** and a second pivot point **44**. Thus, the first shoe assembly **26**, the second shoe assembly **28**, and springs **30** cooperatively allow the robotic tubular stabbing guide **10** to open. The tri-guide design **40** advantageously reduces the opening footprint of the tool, unlike a bifold or a two-piece design. The tri guide design **40** is of a rounded plate shape that is 120 degrees of its total diameter. It includes a first hinge **46** and a second hinge **48** centered on the ends of the plate shape, where the first arm assembly **16**, and the second arm assembly **18** mount. The plate shape of the tri-guide design **40** includes attachments for the springs **30** to mount from the main body assembly **14** to the first arm assembly **16**, and the second arm assembly **18**. The bolt patterns are of a universal type to the arms, which allows the casing and a doper assembly (not shown) to both attach to the assembly. The described assembly is mounted perpendicular to the axis of its diameter for rigidity and allows movement along the rails **36** that the mounting plate **12** slides in between.

[0049] The first arm assembly **16**, and the second arm assembly **18** each make up the other 120 degrees of the total 360 degree diameter and are mirrored for interchangeability. The first arm assembly **16**, and the second arm assembly **18** have bolt patterns, universal to the main body assembly **14**, for mounting the shoe assemblies **26**, **28** and the doper assembly, and also features the pivot points **46**, **48** on one end, where it mounts to the main body assembly **14**, the pivot points where the springs **30** mount, as well as holes for a guide to be installed on the front of each arm.

[0050] The tubular guides **20**, **22**, **24** sizes are 9 $\frac{5}{8}$ to 2 $\frac{3}{8}$ inches and are manufactured from a single cylindrical piece of material, which is cut into three 120-degree segments and features pairs of

holes counterbored universally into each segment, with a metallic sleeve pressed in place to prevent the holes from distorting during use. The casing guide bolt pattern is universal to all guides which facilitates interchangeability in the robotic tubular stabbing guide **10**.

[0051] The casing guides **26, 28** includes geometry that directs the force generated by pressing against a round surface of the casing guides **26, 28** perpendicular to the point of rotation of a tubular, thus allowing the jaws to open when a tubular is pressed against the casing guides **26, 28**. The casing guides **26, 28** force open the arm assemblies **16, 18** and work in pairs.

[0052] The springs **30** are both resistant to cold and corrosion. The springs **30** maintain pressure on the arm assemblies **16, 18** and main body assembly **14**, thus keeping the robotic tubular stabbing guide **10** closed. The springs **30** are each a compression style spring.

[0053] In operation, the robotic tubular stabbing guide **10** includes three steps to its movement. First, the robotic arm while associated with the robotic tubular stabbing guide **10**, positions the robotic tubular stabbing guide **10** above a casing's/pipe's box end, and is lowered until it makes contact with the tubular. Depending on the vertical position of the tubular, the robotic tubular stabbing guide **10** will accommodate some vertical deviation. In the next step, the gas or oil rig lowers a next section of tubular downward, and the tubular guides **20, 22, 24** direct the second casing's/pipe's pin-end to the center of the box-end of the first tubular. The springs **30** keep the first arm assembly **16**, and the second arm assembly **18** of the robotic tubular stabbing guide **10** from opening when the tubular brushes against their sides or is mis-stabbed. If a doper is installed, it would then spray dope along the tubular as it is lowered into position. In a third step, the robotic arm pulls the robotic tubular stabbing guide **10** away from the joint of the first tubular and second tubular and the robotic tubular stabbing guide **10** closes using its springs **30**. An optional fourth step includes pulling the tubular from the well bore. In this optional step, the robotic tubular stabbing guide **10** is pushed onto the tubular via use of the casing guides **26, 28** (shoes) to force open the arm assemblies **16, 18**. The force would open up the guide to a point where the pipe will fit into its center. The robotic tubular stabbing guide **10** will close from the spring tension **30**. Once the tubular joint is backed out or unthreaded the tubular will be lifted out and removed. The robotic tubular stabbing guide **10** will be lifted and removed so the next tubular can be lifted out of the bore hole.

[0054] The robotic tubular stabbing guide **10** is made of 3 materials, including 44 W-50 W steel, 1018 steel, and REDCO 750 polyurethane. It should be appreciated that other suitable materials may be used. The robotic tubular stabbing guide **10**, except the casing guides **20, 22, 24** are made of 44 W-50 W and 1018 steel, due to its material characteristics and availability.

[0055] The casing guide **20, 22, 24** are made of Redco 750 polyurethane. It was chosen because of its abrasion resistance, hardness, temperature rating to -57°C ., along with oil, grease, and chemical resistance.

[0056] The robotic tubular stabbing guide **10** was made with the future of the gas and oil well drilling industry in mind. Considerations were made for adaptability, interchangeability, and ease of assembly. The mounting points universally positioned on the main body and arms allow the rapid design and manufacture of components to complement the robotic tubular stabbing guide **10**. Where possible, off the shelf parts were sourced to reduce lead time and build upon established systems.

[0057] The casing guide likewise uses an easily replicable bolt pattern for installation, thus allowing for the rapid design and installation of casing guides for different diameter casings.

[0058] This robotic tubular stabbing guide **10** is specifically suitable for oil rig use in the conditions: where workers cannot be present on the deck; the hydraulics are saturated; sub-zero climates; a need to allow the precise alignment of casing lengths with each other; and subsequent retraction once the pipe section is connected. The robotic tubular stabbing guide **10** will operate without outside intervention and is resistant to high cycle rates, corrosion, and cold weather while operating.

[0059] With reference to FIGS. **26A-26C**, the robotic tubular stabbing guide **10** may be converted to an open configuration when one or more biasing mechanisms may have force applied via the first shoe assembly **26** and the second shoe assembly **28**.

[0060] With reference to FIGS. **27A-27D**, the robotic tubular stabbing guide **10** may open and close about a tubular. In some embodiments, the robotic tubular stabbing guide **10** may be pressed upon a tubular via a robotic arm.

[0061] For example, the robotic arm may connect to the robotic tubular stabbing guide **10** via the mount **12**. In some embodiments, the robotic tubular stabbing guide **10** makes contact with the tubular via the first shoe assembly **26** and the second shoe assembly **28**, which center the robotic tubular stabbing guide **10** with the tubular and transfer force from the robotic arm pushing the robotic tubular stabbing guide **10** onto the tubular to the springs **30**. In some embodiments, the springs **30** may relent to the force and allow the robotic tubular stabbing guide **10** to convert from a closed configuration and an open configuration. In some embodiments, when converting to the open configuration, the robotic tubular stabbing guide **10** may allow the tubular to slide into the interior cavity of the robotic tubular stabbing guide **10**. In some embodiments, when the robotic tubular stabbing guide **10** has been pressed upon the tubular with an amount of force greater than a threshold, and the tubular is aligned with the interior cavity, the robotic tubular stabbing guide **10** may close around the robotic tubular stabbing guide **10** via the springs **30**.

[0062] FIG. **28** illustrates an example process **2800** for utilizing the robotic tubular stabbing guide **10** to secure a tubular during installation. In some implementations, one or more process blocks of FIG. **28** may be performed by a human operator or one or more processors executing instructions stored on a non-transitory computer-readable storage medium.

[0063] As shown in FIG. **28**, process **2800** may include positioning the robotic tubular stabbing guide **10** via a robotic arm over a coupling fastened to a first tubular installed in a wellbore (block **2802**). For example, a first tubular may be vertically installed in a well bore with a coupling affixed to the top of the first tubular. In some embodiments, the first tubular may be configured for connection via threaded couplings, the tubular having helical grooves or threads cut into the outer surface at one or both ends. In some embodiments, the coupling may have complementary threads cut into the interior for installation with a threaded tubular.

[0064] As also shown in FIG. **28**, process **2800** may include lowering the robotic tubular stabbing guide **10** along the length of the coupling until a bottom of the robotic tubular stabbing guide **10** is aligned with a bottom thread of the first tubular (block **2804**). For example, the robotic tubular stabbing guide **10** may be in a closed configuration with the central cavity centered around the circumference of the coupling where the central cavity is larger than the circumference of the coupling. In some embodiments, the robot arm may lower the robotic tubular stabbing guide **10** so that the coupling is encircled, and the robotic tubular stabbing guide **10** may be lowered along the length of the coupling. In some embodiments, the robotic tubular stabbing guide **10** may be lowered past the coupling to the first tubular and may align the bottom lip of the tri-guide design **40** with the bottommost thread of the first tubular. In some embodiments, the first tubular may have a lip or ring protruding from the surface at the point where the threads start. In some embodiments, the robotic tubular stabbing guide **10** may be lowered until the bottommost point contacts the lip, indicating the start of the threads of the first tubular.

[0065] As further shown in FIG. **28**, process **2800** may include stabbing a second tubular through the top of the robotic stabbing guide **10**, which aligns the second tubular with the coupling (block **2806**). In some embodiments, the robotic tubular stabbing guide **10** may have secured the first tubular and the coupling from moving and twisting with the tri-guide design **40** protruding above the top of the coupling for aligning the second tubular with the coupling. For example, a top drive may lower the second tubular onto the coupling and guide the threads of the second coupling into the reciprocal threads inside the coupling via the tri-guide design **40**.

[0066] As also shown in FIG. **28**, process **2800** may include installing the second tubular with the

coupling (block **2808**). While aligned, the top drive may twist the second tubular into the coupling, creating an airtight seal, while the robotic tubular stabbing guide **10** prevents the coupling and first tubular from twisting. In some embodiments, the polyurethane guide holds the pipe in its 360° circumference so that the pin end contacts and slides along the angled wall of the top portion of the polyurethane guide. Thus, the robotic tubular stabbing guide **10** keeps the tubular threads, shoulders, and seals from being damaged.

[0067] As further shown in FIG. **28**, process **2800** may include removing the robotic tubular stabbing guide **10** from contact with the first tubular, the second tubular, and the coupling (block **2810**). For example, the robotic arm may pull the robotic tubular stabbing guide **10** from contact with the first tubular, the second tubular, and the coupling by moving in a direction perpendicular to the alignment of the first tubular. For example, the tubular may be installed vertically into a wellbore, and the robotic tubular stabbing guide **10** may move horizontally. In some embodiments, the robotic tubular stabbing guide **10** may convert from the closed configuration to the open configuration. This step may operate as a reverse of the process described in FIG. **27**.

[0068] Although FIG. **28** shows example blocks of process **2800**, in some implementations, process **2800** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **28**. Additionally, or alternatively, two or more of the blocks of process **2800** may be performed in parallel.

[0069] FIG. **29** illustrates an example process **2900** for utilizing robotic tubular stabbing guide **10** to secure a tubular during uninstallation. In some implementations, one or more process blocks of FIG. **29** may be performed by a human operator or one or more processors executing instructions stored on a non-transitory computer-readable storage medium.

[0070] As shown in FIG. **29**, process **2900** may include positioning the robotic tubular stabbing guide **10**, via a robotic arm, perpendicularly to a first tubular vertically installed in a well bore, a coupling fixed to the top of the first tubular, and a second tubular fixed to the top of the coupling (block **2902**). For example, the robotic tubular stabbing guide **10** may be positioned so that the first shoe assembly **26** and the second shoe assembly **28** are directed to the side of the first tubular. In some embodiments, the process of engaging the robotic tubular stabbing guide **10** with the first tubular may be similar to the process described in FIG. **27**.

[0071] As also shown in FIG. **29**, process **2900** may include pushing the robotic tubular stabbing guide **10** onto the second tubular and closing the robotic tubular stabbing guide around the second tubular (block **2904**). In some embodiments, the robotic tubular stabbing guide **10** makes contact with the tubular via the first shoe assembly **26** and the second shoe assembly **28**, which center the robotic tubular stabbing guide **10** with the tubular and transfer force from the robotic arm pushing the robotic tubular stabbing guide **10** onto the tubular to the springs **30**. In some embodiments, the springs **30** may relent to the force and allow the robotic tubular stabbing guide **10** to convert from a closed configuration and an open configuration. In some embodiments, when converting to the open configuration, the robotic tubular stabbing guide **10** may allow the tubular to slide into the interior cavity of the robotic tubular stabbing guide **10**. In some embodiments, when the robotic tubular stabbing guide **10** has been fully pressed upon the tubular, the robotic tubular stabbing guide **10** may close around the robotic tubular stabbing guide **10** via the springs **30**, which may cause the robotic tubular stabbing guide **10** to convert back to a closed configuration.

[0072] As further shown in FIG. **29**, process **2900** may include lowering the robotic tubular stabbing guide **10** along the length of the second tubular and the coupling until a bottom rim of the robotic tubular stabbing guide is aligned with a bottom thread of the first tubular (block **2906**). In some embodiments, the robotic tubular stabbing guide **10** may be lowered along the length of the second tubular and the coupling until the first tubular is reached. In some embodiments, the robotic tubular stabbing guide **10** may be lowered past the coupling to the first tubular and may align the bottom lip of the tri-guide design **40** with the bottommost thread of the first tubular. In some embodiments, the tubulars may have a lip or ring protruding from the surface at the point where the

threads start. In some embodiments, the robotic tubular stabbing guide **10** may be lowered until the bottommost point contacts the lip, indicating the start of the threads of the first tubular.

[0073] As also shown in FIG. **29**, process **2900** may include uninstalling the second tubular from the coupling while the robotic tubular stabbing guide **10** secures the coupling and the first tubular (block **2908**). While aligned, the top drive may twist the second tubular out of the coupling, while the robotic tubular stabbing guide **10** secures the coupling and first tubular and protects from wear to the threads and surface of the components.

[0074] As further shown in FIG. **29**, process **2900** may include removing the robotic tubular stabbing guide **10** from contact with the first tubular and the coupling (block **2910**). For example, the robotic arm may pull the robotic tubular stabbing guide **10** from contact with the first tubular and the coupling by moving in a direction perpendicular to the alignment of the first tubular. In some embodiments, the robotic tubular stabbing guide **10** may convert from the closed configuration to the open configuration while the tubular is moved out of the cavity and return to a closed configuration when the tubular is fully removed from contact with the robotic tubular stabbing guide **10**. This step may operate as a reverse of the process described in FIG. **27**.

[0075] Although FIG. **29** shows example blocks of process **2900**, in some implementations, process **2900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **29**. Additionally, or alternatively, two or more of the blocks of process **2900** may be performed in parallel.

CLAUSES

[0076] Example Clause A: A robotic tubular stabbing guide for oil and gas well application, may include: a mount adapted to connect to a robotic arm; a main body associated with the mount, the main body including: a cylindrical tubular guide including a top portion and a bottom portion; a top portion having an inwardly angled surface relative to an axis passing through a center of the cylindrical guide and a bottom portion having an inner surface colinear to the axis; and a first arm and second arm which are in pivoting association with the main body.

[0077] Example Clause B: The robotic tubular stabbing guide of Example Clause A, where the top portion of the cylindrical guide having an inwardly angled surface relative to an axis passing through a center of the cylindrical guide.

[0078] Example Clause C: The robotic tubular stabbing guide of Example Clause A or Example Clause B, where the bottom portion of the cylindrical guide having an inner surface colinear to the axis.

[0079] Example Clause D: The robotic tubular stabbing guide of any one of Example Clauses A-C, where the first arm and the second arm are biased into a closed position by at least one biasing element.

[0080] Example Clause E: The robotic tubular stabbing guide of any one of Example Clauses A-D, where the first arm biases a first section of the cylindrical tubular guide and the second arm biases a second section of the cylindrical tubular guide.

[0081] Example Clause F: The robotic tubular stabbing guide of any one of Example Clauses A-E, where the first arm includes a first shoe portion curved and extending outwardly away from the cylindrical tubular guide.

[0082] Example Clause G: The robotic tubular stabbing guide of any one of Example Clauses A-F, where the second arm includes a second shoe portion curved and extending outwardly away from the cylindrical tubular guide.

[0083] Example Clause H: A method for operating a robotic tubular stabbing guide to install a tubular, the method may include: positioning the robotic tubular stabbing guide, via a robotic arm, over a coupling fastened to a first tubular installed in a well bore; lowering the robotic tubular stabbing guide along the length of the coupling until a bottom of the robotic tubular stabbing guide is aligned with a bottom thread of the first tubular; stabbing a second tubular through the top of the robotic stabbing guide, which aligns the second tubular with the coupling; installing the second

tubular with the coupling; and removing the robotic tubular stabbing guide from contact with the first tubular, the second tubular, and the coupling.

[0084] Example Clause I: The method of Example Clause H, where the robotic tubular stabbing guide protects the threads of the first tubular while installing the second tubular with the coupling.

[0085] Example Clause J: The method of Example Clause H or Example Clause I, further may include removing the robotic tubular stabbing guide by moving the robotic tubular stabbing guide, via the robotic arm, in a direction which is perpendicular with an alignment of the first tubular and the second tubular.

[0086] Example Clause K: The method of any one of Example Clauses H-J, where moving the robotic tubular stabbing guide in the direction causes the robotic stabbing guide to convert from a closed position to an open position while in contact with the first tubular, the second tubular, and the coupling.

[0087] Example Clause L: The method of any one of Example Clauses H-K, where moving the robotic tubular stabbing guide in the direction until no longer in contact with the first tubular, the second tubular, and the coupling causes the robotic stabbing guide to convert from the open position to the closed position.

[0088] Example Clause M: The method of any one of Example Clauses H-L, further may include guiding of the second tubular, an inwardly angled surface on an interior rim of the robotic tubular stabbing guide, where the inwardly angled surface is relative to an axis passing through a center of the robotic tubular stabbing guide.

[0089] Example Clause N: The method of any one of Example Clauses H-M, where the robotic tubular stabbing guide is in a closed position, where the closed position is maintained by at least one biasing element.

[0090] Example Clause O: A method for operating a robotic tubular stabbing guide to uninstall a tubular, the method may include: positioning the robotic tubular stabbing guide, via a robotic arm, perpendicularly to a first tubular vertically installed in a well bore, a coupling fixed to the top of the first tubular, and a second tubular fixed to the top of the coupling; pushing the robotic tubular stabbing guide onto the second tubular and closing the robotic tubular stabbing guide around the second tubular; lowering the robotic tubular stabbing guide along the length of the second tubular and the coupling until a bottom rim of the robotic tubular stabbing guide is aligned with a bottom thread of the first tubular; uninstalling the second tubular from the coupling while the robotic tubular stabbing guide secures the coupling and the first tubular; and removing the robotic tubular stabbing guide from contact with the first tubular and the coupling.

[0091] Example Clause P: The method of Example Clause O, where the robotic tubular stabbing guide protects the threads of the first tubular while uninstalling the second tubular from the coupling.

[0092] Example Clause Q: The method of Example Clause O or Example Clause P, where pushing the robotic tubular stabbing guide on the second installed tubular causes the robotic stabbing guide to convert from a closed position to an open position.

[0093] Example Clause R: The method of any one of Example Clauses O-Q, where closing the robotic tubular stabbing guide around the second installed tubular, causes the robotic stabbing guide to convert from the open position to the closed position via at least one biasing element.

[0094] Example Clause S: The method of any one of Example Clauses O-R, where pushing the robotic tubular stabbing guide onto the second installed tubular further may include a wing of the robotic tubular stabbing guide contacting the second tubular to inform the conversion of the robotic tubular stabbing guide to the open position.

[0095] Example Clause T: The method of any one of Example Clauses O-S, where securing the coupling and the first tubular includes preventing any rotation.

[0096] It should be appreciated that the robot and components thereof are not a part of the invention and are well known to persons of ordinary skill in the art. The invention and any claims

to the invention are not intended to include the robot or any part thereof as a claim element in the claims. The same is true with the casings/pipes which are well known in the prior art and not intended to be a part of the invention or a required claim element in the claims.

[0097] Several embodiments have been discussed in the foregoing description. However, the embodiments discussed herein are not intended to be exhaustive or limit the invention to any particular form. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the invention may be practiced otherwise than as specifically described.

Claims

1. A robotic tubular stabbing guide for oil and gas well application, comprising: a mount adapted to connect to a robotic arm; a main body associated with the mount, the main body including: a cylindrical tubular guide including a top portion and a bottom portion; a top portion having an inwardly angled surface relative to an axis passing through a center of the cylindrical guide and a bottom portion having an inner surface colinear to the axis; and a first arm and second arm which are in pivoting association with the main body.
2. The robotic tubular stabbing guide of claim 1, wherein the top portion of the cylindrical guide having an inwardly angled surface relative to an axis passing through a center of the cylindrical guide.
3. The robotic tubular stabbing guide of claim 2, wherein the bottom portion of the cylindrical guide having an inner surface colinear to the axis.
4. The robotic tubular stabbing guide of claim 1, wherein the first arm and the second arm are biased into a closed position by at least one biasing element.
5. The robotic tubular stabbing guide of claim 4, wherein the first arm biases a first section of the cylindrical tubular guide and the second arm biases a second section of the cylindrical tubular guide.
6. The robotic tubular stabbing guide of claim 5, wherein the first arm includes a first shoe portion curved and extending outwardly away from the cylindrical tubular guide.
7. The robotic tubular stabbing guide of claim 5, wherein the second arm includes a second shoe portion curved and extending outwardly away from the cylindrical tubular guide.
8. A method for operating a robotic tubular stabbing guide to install a tubular, the method comprising: positioning the robotic tubular stabbing guide, via a robotic arm, over a coupling fastened to a first tubular installed in a well bore; lowering the robotic tubular stabbing guide along the length of the coupling until a bottom of the robotic tubular stabbing guide is aligned with a bottom thread of the first tubular; stabbing a second tubular through the top of the robotic stabbing guide, which aligns the second tubular with the coupling; installing the second tubular with the coupling; and removing the robotic tubular stabbing guide from contact with the first tubular, the second tubular, and the coupling.
9. The method of claim 8, wherein the robotic tubular stabbing guide protects the threads of the first tubular while installing the second tubular with the coupling.
10. The method of claim 8, further comprising removing the robotic tubular stabbing guide by moving the robotic tubular stabbing guide, via the robotic arm, in a direction which is perpendicular with an alignment of the first tubular and the second tubular.
11. The method of claim 10, wherein moving the robotic tubular stabbing guide in the direction causes the robotic stabbing guide to convert from a closed position to an open position while in contact with the first tubular, the second tubular, and the coupling.
12. The method of claim 11, wherein moving the robotic tubular stabbing guide in the direction until no longer in contact with the first tubular, the second tubular, and the coupling causes the robotic stabbing guide to convert from the open position to the closed position.

- 13.** The method of claim 8, further comprising guiding of the second tubular, an inwardly angled surface on an interior rim of the robotic tubular stabbing guide, wherein the inwardly angled surface is relative to an axis passing through a center of the robotic tubular stabbing guide.
- 14.** The method of claim 8, wherein the robotic tubular stabbing guide is in a closed position, wherein the closed position is maintained by at least one biasing element.
- 15.** A method for operating a robotic tubular stabbing guide to uninstall a tubular, the method comprising: positioning the robotic tubular stabbing guide, via a robotic arm, perpendicularly to a first tubular vertically installed in a well bore, a coupling fixed to the top of the first tubular, and a second tubular fixed to the top of the coupling; pushing the robotic tubular stabbing guide onto the second tubular and closing the robotic tubular stabbing guide around the second tubular; lowering the robotic tubular stabbing guide along the length of the second tubular and the coupling until a bottom rim of the robotic tubular stabbing guide is aligned with a bottom thread of the first tubular; uninstalling the second tubular from the coupling while the robotic tubular stabbing guide secures the coupling and the first tubular; and removing the robotic tubular stabbing guide from contact with the first tubular and the coupling.
- 16.** The method of claim 15, wherein the robotic tubular stabbing guide protects the threads of the first tubular while uninstalling the second tubular from the coupling.
- 17.** The method of claim 15, wherein pushing the robotic tubular stabbing guide on the second installed tubular causes the robotic stabbing guide to convert from a closed position to an open position.
- 18.** The method of claim 17, wherein closing the robotic tubular stabbing guide around the second installed tubular, causes the robotic stabbing guide to convert from the open position to the closed position via at least one biasing element.
- 19.** The method of claim 17, wherein pushing the robotic tubular stabbing guide onto the second installed tubular further comprises a wing of the robotic tubular stabbing guide contacting the second tubular to inform the conversion of the robotic tubular stabbing guide to the open position.
- 20.** The method of claim 15, wherein securing the coupling and the first tubular includes preventing any rotation.
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